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Course Code: IT A0501 - ~~Mechine~~
Computer Vision

Assessment: Co2_AT3_Class test

Grey-level transformation for a Dark Museum photograph

Concept:- Gamma transformation improves the Brightness of dark images and reveals details in shadow regions.

Formula:-

$$S = Cr^{\gamma}$$

r is the Input intensity, S is the Output intensity C is a Constant and γ is the gamma Value.

Steps:-

1) Read the dark photograph and Convert it to grayscale if required.

2) Normalize pixel intensities to the range 0-1

3) Select $\gamma < 1$ such as 0.5, to Brighten dark regions.

4) Apply the transformation to all pixels and scale the result to 0-255.

5. Display the enhanced image

Result:- Dark regions become brighter, revea
hidden exhibit details while preserving the o

image structure

2) Histogram Equalization for Fingerprint

Enhancement

Concept:- Histogram equalization improves contras
by redistributing image intensity values.

Steps:-

1) Read the fingerprint image and convert it to
grayscale.

2) Calculate the Histogram of pixel intensities.

3) Normalize the Histogram to obtain pixel probabilities

4) Calculate the cumulative distribution function
(CDF) and map the original intensities to new values

mula:

$$S_K = (L-1) \sum_{j=0}^{K} p(r_j)$$

L is the Number of intensity Value levels

$p(r_j)$ is the Normalized histogram

5) Replace the original pixel Values with the mapped Values.

Result:- Fingerprint ridges and valleys become clearer, improving Contrast and supporting fingerprint identification.

2) Salt and-pepper Noise Removal using a Median filter.

Concept:-

A Median filter removes salt and pepper noise while preserving edges better than a averaging filter

Steps

1) Read the noisy image

2) Select 3×3 neighborhood around each p

3) Arrange the neighborhood values in ascend

order

4) Replace the central pixel with the median val

5) Repeat for all pixels to obtain the filtered im

Justification:- An average filter is affected by extreme black and white noise values and may blur edges. A Median filter rejects these extreme values more effectively.

Result:-

Noise is reduced while important object boundaries and details remain comparatively sharp.

4) PCB track Enhancement using Unsharp masking

Concept:- Unsharp masking sharpens a blurred image

dding enhanced high frequency details to the original image.

Formula:-

$$g(x,y) = f(x,y) + k[f(x,y) - f_b(x,y)]$$

* f is the original image
* f_b is the blurred image
* k controls sharpening strength

Steps:-

1) Read the blurred printed circuit board image
2) Apply Gaussian blur to obtain a smoothed image
3) Subtract the smoothed image from the original to create a detail mask.
4) Add the scaled detail mask to the original image
5) Adjust k and display the sharpened image.

Result: Copper tracks becomes sharper and to inspect for defects. Excessive sharpening should be avoided because it may amplify & create unwanted edges.

5.) Draw Image smoothing using a Gaussian low-pass filter.

Concept:-

A frequency-domain low pass filter reduces high-frequency noise while retaining the main image structure.

Formula:-

$$H(u,v) = e^{-D(u,v)^2/(2D_0^2)}$$

* $D(u,v)$ is the distance from the frequency origin
* D_0 is the cutoff frequency.

1)- Convert the drone image to grayscale

2) Apply the Fourier transform to convert it into the frequency domain.

3) Construct a Gaussian low-pass filter

4) Multiply the Fourier transformed image by the filter.

5) Apply the Inverse fourier transform to obtain the smoothed image.

Result:-

Grainy noise is reduced, improving the visibility of farm-plot outlines. A low cutoff frequency produce stronger smoothing but remove more details, whereas high cutoff preserves more details but allow more noise to remain.

1) Retinal Blood vessel Enhancement using Filtering.

Concept:- High boost filtering enhances fine and edges while retaining the original image information.

Formula:-

$$g(x,y) = Af(x,y) - f_b(x,y), \quad A > 1$$

* f is the original image
* f_b is the blurred image
* A controls the enhancement strength

Steps:-

1) Read the Retinal fundus image and convert it to grayscale.
2) Apply a low pass filter to produce a blurred image.

3) Subtract the Blurred image from the original to obtain fine details.

4) Add the amplified detail information to the Original RGB image.

5) Adjust HSV and display the enhanced image

Result:-

Thin Blood Vessels and fine retinal structures become more visible against the background.

The enhancement level must be controlled to avoid amplifying noise or hiding small vessels.

7) point and line detection for Batik defects

Concept:-

point detection identifies isolated intensity changes.

line
While detection identifies thin structures such as cracks.

point detection mask:

$$W_p = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Horizontal line detection mask:

$$W_{h2} \begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

Steps:

1) Convert the bottle image to grayscale and reduce noise

2) Apply the point detection mask to identify possible air bubbles.

3) Apply Horizontal, Vertical, and diagonal line masks to identify cracks.

4) Calculate the absolute response of each crack

5) Apply a suitable threshold to mask significant

spons as defects

Result :-

Isolated bubbles and Hairline cracks are can be Identified. A low Threshold may detect noise as defects. Whereas a high threshold may miss small defects. The threshold should be selected using background or noise levels and Validated defect samples.

8) Handwriting Extraction using Adaptive Thresholding.

Concept :- Adaptive thresholding separates dark handwriting from the background using local intensity variations.

Steps :-

1) Read the answer sheet image and Convert it to

grayscale.

2) Apply Gaussian blur to reduce noise

3) Divide the Image into small neighbo

4) Calculate a local Threshold using the ad
mean or Gaussian method.

5) Classify pixels darker than the local Thre
as ink and the remaining pixels as background.

For Comparsion, global thresholding use a singl
threshold T:

$$g(x,y) = \begin{cases} 0, & f(x,y) < T \\ 255, & f(x,y) \geq T \end{cases}$$

Justification:- Global thresholding may fail when illumination varies across the sheet. Adaptive thresholding calculates local thresholds and handles uneven lighting more effectively.

sult:-

Handwritten characters are separated from the paper background, improving readability and supporting optical character recognition.

9) Lake Extraction using Region Recognition

Concept:-

Region growing extract a connected region by adding neighbouring pixels that satisfy a similarity criterion.

Steps:-

1) Read the satellite Image and select a seed point inside the lake.

2) Determine the seed pixels intensity or color

3) Examine Neighbouring pixels and compare their values with the seed:

4) Add pixels satisfying the condition:

$$|I(x,y) - I_s| \leq T$$

* $I(x,y)$ is the neighbouring pixel intensity
* I_s is the seed intensity
* T is the tolerance

5) Continue growing the region until no more neighbouring pixels satisfy the condition.

6) Display the extracted lake as a segmented mask.

Result:-

The lake is separated from the surrounding land based on pixel similarity. The seed should be selected inside the lake, and the tolerance should be adjusted to include water variations without merging the surrounding land.

Edge linking and Boundary Detection for Mechanical Gears

Concept:-

Edges linking connects fragmented edge pixels, while boundary detection extracts a complete outline of an object.

Steps:-

1) Convert the gear image to grayscale and reduce noise

2) Apply the Canny edge detector to identify initial edges

3) link nearby edges pixels using distance, gradient direction, and orientation similarity.

4) Apply a Hough transform to detect suitable geometric edge structures.

5) Use Morphological closing to fill small gaps, if required.

6. Extract Connected Contours and select the Contour Corresponding to the gear boundary.

Result:-

Broken edge segments caused by shadows and noise are connected to produce a more complete gear outline. proper thresholds and contour selection help avoid false boundaries and improve mechanical Inspection.